

Junxian Ma

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RESEARCH INTERESTS

Focus: Distributed LLM training and inference, sparse attention, GPU kernel optimization, and algorithm-system co-design

EDUCATION

Tsinghua University

B.Eng. Candidate in Computer Science and Technology | Beijing, China

2024 - 2028 (Expected)

INDUSTRY EXPERIENCE

Zhipu AI | Multimodal Algorithm Research Intern

GLM-4.6V Series | Beijing, China

Oct 2025 - Feb 2026

- Contributed to multimodal alignment and long-video understanding for the GLM-4.6V series.
- Used slime and SGLang in the reinforcement-learning training pipeline for GLM-4.6V.
- Built large-scale multimodal data pipelines for cleaning, filtering, and quality control.

SELECTED RESEARCH AND SYSTEMS PROJECTS

FusedIndexTopK | High-Performance Indexer + Top-K Fusion for DeepSeek Sparse Attention | [GitHub](#)

Jul 2026 - Present

Independent GPU Systems Research | DeepSeek-V3.2 | NVIDIA H20

- Built an exact fused operator for the Indexer+Top-K stage in DeepSeek-V3.2 sparse attention, replacing the conventional two-stage pipeline that materializes and rereads a dense FP32 score matrix in HBM.
- Co-designed a bounded-memory pipeline combining DeepGEMM-style FP8 score production, compact candidate generation, exact radix selection, and device-side underflow/overflow repair.
- Outperformed DeepGEMM+FlashInfer in all 40 held-out H20 workload cells; median end-to-end latency reductions were 14.7% versus DeepGEMM+FlashInfer (95% CI: 13.2-15.9%) and 30.1% versus DeepGEMM+torch.topk.
- Extended exact selection via fixed-memory 16K chunking and online Top-2,048 merging, retaining 4.9-7.8% gains at 32K-128K in development evaluation; released correctness tests, methodology, and machine-readable results.

Sparse-Attention Algorithm-Infra Co-design | [NSA-Exp](#)

2026

Independent Systems Analysis and H100 Experiments

Writing: [Sparse Attention: Algorithm-Infra Co-design](#)

- Compared NSA and DeepSeek-V3.2 DSA from token selection to GPU execution, identifying how block layout, head sharing, and memory movement determine whether theoretical sparsity becomes real speedup.
- Profiled the public DSA pipeline from 32K to 2M context on H100. Discovery (Indexer+Top-K) already exceeded FlashMLA at the 64K checkpoint and widened rapidly with context length; eight randomized row-order tests stayed within 0.47% of sorted access, indicating that finding tokens, rather than address ordering, dominated the tested setup.

ADDITIONAL TECHNICAL EXPERIENCE

Tsinghua University Supercomputing Team | Training Participant

2026 - Present

- Training alongside the team in parallel programming, GPU/HPC optimization, and performance profiling.

Tsinghua University | Quickviewer Multimodal LLM

Feb 2025 - Aug 2025

Academic Rising Stars Program | Department of Computer Science

- Ran multimodal data preprocessing, vision-encoder alignment, and momentum-based chunking experiments for visual question answering and fine-grained long-video understanding.

Open-Cowork | Open-Source Desktop AI Agent | [GitHub](#)

2026

- Built Electron/React/TypeScript features and maintained Skills, document-generation, and WeChat integration workflows across Windows and macOS.

TECHNICAL SKILLS

Programming: Python, C/C++, CUDA, Triton, TypeScript

ML and Systems: PyTorch, slime, SGLang, DeepGEMM, FlashInfer, FlashMLA, FLA, Nsight Systems/Compute, Kineto/CUPTI, Linux, Git

SELECTED HONORS

2024 Ranked 1st in Gansu Province, National College Entrance Examination

2023 National Bronze Medal, 40th Chinese Physics Olympiad Final

2025 Outstanding Student Leader, Tsinghua University

2024 Freshman First-Class Scholarship, Tsinghua University